

9 (short)

In salt (sodium chloride NaCl), Cl ions form a FCC lattice, and Na ions occupy all the available octahedral holes. The crystal structure of NaCl is shown in Fig. 9.

(a) NaCl has a lattice constant $a = 0.564$ nm. Determine the density of NaCl given that the atomic masses of sodium and chlorine are 22.989 and 35.453 kg/kmol, respectively. [5]

(b) The ratio of the diameters of sodium and chlorine ions is 0.69. Show that the chlorine and sodium ions are touching along the edges of the unit cell. Find the size (as a fraction of the diameter of the Cl ions) of the gaps between the chlorine ions on the diagonal of the face of the unit cell. [5]

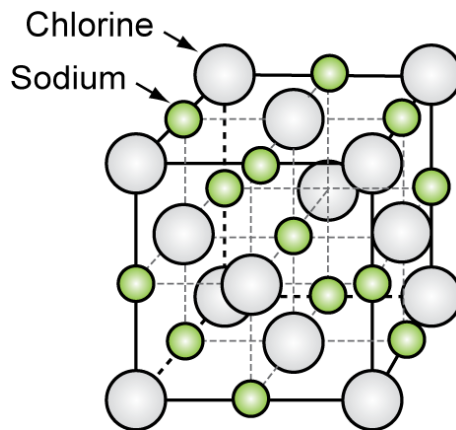


Fig. 9